# Log-periodic oscillation in the all-heads coin game

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**Abstract**

We study the large-$n$ behaviour of the all-heads coin game below the fair coin. Two phenomena are established. First, for every $p \in (0, \frac{1}{2})$ the value $b_{n,p}$ of the greedy strategy All is asymptotically *log-periodic*: it is asymptotic to a continuous, non-constant function of $\log n$ with period $\log\bigl(1/(1-p)\bigr)$, and in particular does not converge. Second — and this is the main result — the *optimal* winning probability $w_{n,p}$ inherits this oscillation: for every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually monotone and has infinitely many strict local maxima and minima. The proof reduces the statement to two one-sided assertions, treats the increasing side by an elementary Bellman estimate, and the decreasing side by an exact transfer to strategy All followed by a renormalisation argument that propagates the oscillation across geometric scales. This paper is the asymptotic companion to the exact and perturbative analysis of [1].

*Working draft. Steps still to be completed are flagged in the text by boxed* **OPEN GAP** *markers (Gaps A–D); Section 9 consolidates their status.*

## 1 Introduction

### 1.1 The game

Fix $p \in (0, 1)$ and write $q := 1 - p$. A player holds $n$ coins, each of which lands heads independently with probability $p$. In each round all remaining coins are tossed; the player must set aside at least one coin showing heads, and loses if no coin shows heads. The game is won once every coin has been set aside. Writing $w_{n,p}$ for the optimal winning probability, the dynamic-programming principle gives the Bellman equation

$$w_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \le m \le n-1} w_{m,p}, \qquad w_{0,p} := 1. \tag{1}$$

Two strategies are natural: One sets aside a single head each round (value $a_{n,p}$), and All sets aside *every* head. The winning probability $b_{n,p}$ of strategy All is defined by the linear recursion

$$b_{n,p} = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j \, b_{j,p}, \qquad b_{0,p} := 1, \tag{2}$$

obtained by conditioning on the number $j$ of coins showing tails in the first round: those $j$ coins are re-tossed, while the $n - j \ge 1$ heads are set aside for good (the term $j = 0$ has weight $p^n$ and wins immediately). The companion paper [1] determines $w_{n,p}$ *exactly* at $p = \frac{1}{2}$, identifies the limit $W(p) = \lim_n w_{n,p}$ for $p > \frac{1}{2}$, and gives a first-order perturbation expansion for $p < \frac{1}{2}$ near $\frac{1}{2}$. All of those results are machine-checked in Lean 4.

The present paper concerns the regime left open by a perturbative analysis: the genuine large-$n$ behaviour for $p < \frac{1}{2}$, which turns out to be *oscillatory* rather than convergent.

## 1.2 Main results

We state the two theorems at once; everything else in the paper is in their service.

**Theorem 1.1** (oscillation of the optimal value)**.** *For every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually monotone. It has infinitely many strict local maxima and infinitely many strict local minima.*

**Theorem 1.2** (log-periodicity of strategy ALL)**.** *For every $p \in (0, \frac{1}{2})$ there is a continuous, strictly positive, non-constant function $\Phi_p : \mathbb{R} \to \mathbb{R}$, periodic with period $L := \log(1/q)$, such that*

$$b_{n,p} = \Phi_p(\log n)\,\bigl(1 + o(1)\bigr) \qquad (n \to \infty).$$

*In particular $b_{n,p}$ does not converge:* $\liminf_n b_{n,p} < \limsup_n b_{n,p}$.

Theorem 1.1 sharpens, and corrects the natural first guess about, the first-order picture of [1]: that analysis shows $w_{n,p}$ to be eventually increasing *at first order in* $\frac{1}{2} - p$, but the exact sequence is not eventually monotone at all. The oscillation is non-perturbative — its amplitude is exponentially small in $1/(\frac{1}{2} - p)$ — which is why it is invisible to any finite-order expansion.

## 1.3 Strategy of proof

The proof of Theorem 1.1 has a rigid skeleton, which we record now so that the role of each later section is clear. For a real sequence $(x_n)$ put $d_n := x_{n+1} - x_n$.

> **Step 0 (reduction).** A bounded sequence has infinitely many strict local maxima and minima as soon as it is *not eventually monotone*, i.e. as soon as $d_n > 0$ infinitely often *and* $d_n < 0$ infinitely often (Lemma 5.1). Thus Theorem 1.1 splits into
>
> (A) $(w_{n,p})$ is not eventually non-decreasing, (B) $(w_{n,p})$ is not eventually non-increasing.
>
> **Step A (the increasing side).** On a stretch where $w$ is non-decreasing, the suffix-maximum in (1) collapses to its right endpoint and the Bellman equation becomes the linear strategy-ONE recursion, whose one-step increment has the sign-definite dominant term $-q^n w_{n-1}$. Since $q > p$, this forces an eventual decrease. Statement (A) follows (Proposition 4.1); the argument is elementary and complete.
>
> **Step B (the decreasing side).** On a stretch where $w$ is non-increasing, the suffix-maximum collapses to its *left* endpoint, and $w$ becomes — exactly — a strategy-ALL sequence (Lemma 5.2). Strategy ALL oscillates (Theorem 1.2); hence $w$ cannot have a non-increasing tail. The transfer of the oscillation from ALL to $w$ is carried out by a renormalisation argument (Section 6): a windowed fundamental coefficient $A(t)$ is shown to satisfy a propagation identity across the geometric scale, from which non-convergence — hence non-monotonicity — follows.

The asymmetry between Steps A and B is structural: the increasing side reduces to a contractive linear recursion with a sign-definite drift, the decreasing side to a scale-invariant recursion whose solutions oscillate. Step A is elementary; Step B carries the analytic weight of the paper.

## 2 Setup and notation

Throughout, $p \in (0, \frac{1}{2})$ is fixed, $q = 1-p \in (\frac{1}{2}, 1)$, and we use freely that $q > p$, $p/q < 1$ and $2p < 1$. We write $L := \log(1/q) > 0$ and $\omega := 2\pi/L$.

The value $w_{n,p}$ satisfies (1); it is a probability, so $0 < w_{n,p} \le 1$, and in fact $w_{n,p} < 1$ for $n \ge 1$. The two reference strategies are dominated by the optimum. Strategy All has winning probability $b_{n,p}$, defined by the recursion (2); strategy One satisfies the linear recursion

$$a_{n,p} = p^n + (1 - p^n - q^n)\, a_{n-1,p}, \qquad a_{0,p} := 1. \tag{3}$$

In both cases $w_{n,p} \ge \max(a_{n,p}, b_{n,p})$, since the optimum dominates any fixed strategy. We use the binomial identities $\sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - q^n$ and $\sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - q^n$. Where the value of $p$ is fixed we abbreviate $w_n := w_{n,p}$, $b_n := b_{n,p}$.

## 3 Strategy All: log-periodic asymptotics

This section proves Theorem 1.2. We summarise the analysis; the full development is in the companion note `strategy_all_asymptotics.tex`.

### 3.1 The functional equation

Let $B(z) := \sum_{n \ge 0} b_n z^n / n!$ be the exponential generating function of strategy All. The convolution structure of (2) yields the first-order $q$-difference equation

$$B(z) = 1 + (e^{pz} - 1)\, B(qz), \tag{4}$$

and iterating it (the remainder tends to 0 since each factor $e^{pq^k z} - 1 \to 0$),

$$B(z) = \sum_{M \ge 0} \prod_{k=0}^{M-1} \left(e^{pq^k z} - 1\right). \tag{5}$$

Extracting coefficients exhibits $b_n/p^n$ as a $q$-analogue of the Fubini (ordered-Bell) polynomials; equivalently, with $X_1, \dots, X_n$ i.i.d. geometric, $b_{n,p} = \Pr(\{X_1, \dots, X_n\} = \{1, \dots, \max_i X_i\})$.

### 3.2 Saddle-point and Mellin analysis

Writing $e^{pq^k z} - 1 = e^{pq^k z}(1 - e^{-pq^k z})$ and using $p \sum_{k=0}^{M-1} q^k = 1 - q^M$,

$$\prod_{k=0}^{M-1} (e^{pq^k z} - 1) = e^{\,z(1-q^M)} \prod_{k=0}^{M-1} \left(1 - e^{-pq^k z}\right).$$

On the positive axis the terms of (5) are dominated by $M$ near $K(z) := \log(pz)/\log(1/q)$; a saddle-point evaluation gives $B(z) = e^z\, \Phi(\log z)(1 + o(1))$ with $\Phi$ bounded and periodic of period $L$, and a Hayman-type transfer to coefficients yields $b_{n,p} = \Phi_p(\log n)(1 + o(1))$. A Mellin analysis of the harmonic sum $\sum_k \log(1 - e^{-pq^k z})$ identifies the Fourier coefficients of $\Phi_p$ at frequencies $s_\ell = 2\pi i \ell / L$ as

$$c_\ell = -\frac{1}{L}\, p^{-s_\ell}\, \Gamma(s_\ell)\, \zeta(s_\ell + 1), \qquad \ell \in \mathbb{Z}.$$

For $\ell \ne 0$ these are non-zero: $\Gamma$ has no zeros and $\zeta(1 + it) \ne 0$ for real $t \ne 0$ (the non-vanishing of $\zeta$ on the line $\Re s = 1$). Hence $\Phi_p$ is non-constant and $b_{n,p}$ does not converge — which is Theorem 1.2.

**▶ OPEN GAP — Gap A — Hayman regularity. Work continues here.**

The Hayman-type transfer from $B(z) \sim e^z \Phi(\log z)$ to the coefficient asymptotics $b_n \sim \Phi_p(\log n)$ requires admissibility / sector control of $B(z)$. Here $B$ is entire of order 1 and type 1 (indeed $B(r) \le e^r$, since $0 \le b_n \le 1$); the delicate feature is the non-constant log-periodic factor, which the closure theory of H-admissible functions does not cover. A direct sector saddle-point estimate is needed.

**▶ OPEN GAP — Gap B — Mellin contour. Work continues here.**

The Mellin step — the strip of analyticity and the contour shift for the transform of $\log B(z) - z$ — is standard in spirit (Flajolet–Sedgewick) but is not carried out here.

**Remark 3.1.** Gaps A and B concern the *quantitative* asymptotics. The qualitative conclusion of Theorem 1.2 that we actually use downstream — that $b_{n,p}$ does not converge, equivalently $\Phi_p$ non-constant — is, granted the Mellin framework, secured by $\zeta(1+it) \neq 0$.

# 4 Step A: $w_{n,p}$ is not eventually increasing

**Proposition 4.1.** *For $p \in (0, \frac{1}{2})$ there is no $M$ with $w_n \ge w_{n-1}$ for all $n > M$. That is, $w_n < w_{n-1}$ for infinitely many $n$.*

*Proof.* Suppose $w_n \ge w_{n-1}$ for all $n > M$, i.e. $w$ is non-decreasing on $\{M, M+1, \dots\}$. Fix $n > M+1$ and split the Bellman sum (1) at $j = M$, writing $S_{j,n} := \max_{j \le m \le n-1} w_m$. For $j \ge M$ the index window lies in the non-decreasing range, so $S_{j,n} = w_{n-1}$; for every $j$ one has $w_{n-1} \le S_{j,n} \le 1$. With $U_n := \sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j$ and $T_n := \sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j S_{j,n}$,

$$w_n = p^n + T_n + (1 - p^n - q^n - U_n)\, w_{n-1},$$

hence

$$w_n - w_{n-1} = p^n(1 - w_{n-1}) - q^n w_{n-1} + (T_n - U_n w_{n-1}).$$

Since $w_{n-1} \le S_{j,n} \le 1$ one has $0 \le T_n - U_n w_{n-1} \le U_n$, and $U_n \le (M-1)\, n^{M-1} p^{n-M+1} = C_1\, n^{M-1} p^n$ with $C_1 = (M-1)p^{1-M}$. With $c := w_M > 0$ and $w_{n-1} \ge c$,

$$w_n - w_{n-1} \le \bigl(1 + C_1 n^{M-1}\bigr) p^n - c\, q^n.$$

As $p/q < 1$, the ratio $(1 + C_1 n^{M-1})(p/q)^n \to 0$, so the right-hand side is negative for all large $n$ — contradicting $w_n \ge w_{n-1}$. ☐

Proposition 4.1 is statement (A) of Section 1.3 and is fully rigorous.

# 5 Step B, part 1: reduction and the transfer lemma

**Lemma 5.1** (reduction). *A bounded real sequence $(x_n)$ that is neither eventually non-decreasing nor eventually non-increasing has infinitely many strict local maxima and infinitely many strict local minima, provided $x_n \neq x_{n+1}$ for all large $n$.*

*Proof.* Not eventually non-decreasing means $d_n := x_{n+1} - x_n < 0$ infinitely often; not eventually non-increasing means $d_n > 0$ infinitely often. Between a sign change of $d$ from $-$ to $+$ lies a local minimum, between $+$ and $-$ a local maximum; infinitely many sign changes yield infinitely many of each, strict under $x_n \neq x_{n+1}$. ☐

**Lemma 5.2** (transfer to strategy All)**.** *Suppose $w_n \geq w_{n+1}$ for all $n \geq M$. Define $x_0 := 1$, $x_j := \max_{j\leq m\leq M} w_m$ for $1 \leq j \leq M-1$, and $x_j := w_j$ for $j \geq M$. Then $x_n = w_n$ for $n \geq M$, and $(x_n)$ satisfies the exact strategy-*All *recursion*

$$x_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j x_j \qquad \text{for every } n > M.$$

*Proof.* Fix $n > M$. In the Bellman equation (1) the suffix-maximum $S_{j,n} = \max_{j\leq m\leq n-1} w_m$ collapses to the *left* endpoint on the non-increasing range: for $M \leq j \leq n-1$, $S_{j,n} = w_j = x_j$; for $1 \leq j \leq M-1$, $S_{j,n} = \max(\max_{j\leq m\leq M-1} w_m,\, w_M) = \max_{j\leq m\leq M} w_m = x_j$, independent of $n$. Writing the $p^n$ term as $\binom{n}{0} p^n q^0 x_0$ gives the displayed recursion. □

Lemma 5.2 is the crux of Step B: it says that a non-increasing tail of $w$ is, exactly, a strategy-All sequence with finitely many initial values modified. The remaining task is to show such a sequence cannot exist, by transferring the oscillation of Theorem 1.2.

# 6 Step B, part 2: the renormalisation argument

Throughout this section $(x_n)$ is a bounded solution of the strategy-All recursion for $n > M$ — either $b$ itself, or a transferred sequence of Lemma 5.2.

## 6.1 The recursion as a smoothing in log-scale

With $J_n \sim \mathrm{Bin}(n, q)$ the recursion reads $x_n = E[x_{J_n}]/(1 + q^n)$. Passing to $t := \log n$ and the log-linear interpolation $y$ of $n \mapsto x_n$, and using $\log(qn) = \log n - L$, one application of the recursion shifts the log-scale down by exactly one period $L$ and smooths by a kernel of width $\sigma(t) \sim \sqrt{p/q}\, e^{-t/2} \to 0$.

## 6.2 The local fundamental coefficient

Fix a window $\chi \in C_c^\infty(\mathbb{R})$ with $\int \chi = 1$ and $\hat{\chi}(2\pi) = 0$, and set

$$A(t) := \frac{1}{L} \int_{\mathbb{R}} y(s)\, e^{-i\omega s}\, \chi\!\left(\tfrac{s-t}{L}\right) ds. \tag{6}$$

The condition $\hat{\chi}(2\pi) = 0$ makes $A$ annihilate the constant (mean-level) component, so $A$ extracts the fundamental oscillation. If $x_n$ converges then $A(t) \to 0$.

## 6.3 The propagation estimate

**Proposition 6.1** (propagation)**.** *There are $\kappa(t) \in \mathbb{C}$ and a remainder $r(t)$ with*

$$A(t+L) = \kappa(t)\, A(t) + r(t), \qquad |\kappa(t) - 1| = O(e^{-t}), \qquad |r(t)| = O(e^{-t/2}).$$

The factor $\kappa$ is fully under control.

**Lemma 6.2** (the factor $\kappa$)**.** *For $J_n \sim \mathrm{Bin}(n, q)$, $\kappa(n) = E[(J_n/(qn))^{i\omega} \mathbf{1}_{J_n\geq 1}] = 1 - \frac{p}{2qn}(\omega^2 + i\omega) + O(n^{-3/2})$. In particular $|\kappa - 1| = O(1/n) = O(e^{-t})$.*

*Proof (sketch).* Put $V_n = J_n/(qn) - 1$; the binomial central moments give $E[V_n^2] = p/(qn)$ and $E[|V_n|^3] = O(n^{-3/2})$. Expanding $(1 + V_n)^{i\omega}$ to second order on $\{|V_n| \leq \frac{1}{2}\}$ and bounding the tail by Hoeffding yields the claim. □

For the remainder we record the structural identities on which the estimate rests. Both are exact.

**Lemma 6.3** (windowed-smoothing identities)**.** *Let* $A[y]$ *be given by* (6)*.*

(a) $A[y] \in C^\infty$ *in* $t$*, with* $\partial_t A[y] = -\frac{1}{L} A^{[\chi']}[y]$*; since* $\widehat{\chi'}(2\pi) = 2\pi i\, \hat{\chi}(2\pi) = 0$*, every* $t$*-derivative of* $A[y]$ *is again a constant-annihilating windowed coefficient, hence bounded.*

(b) *For a probability law* $\nu$ *and* $S_\nu y(s) := \int y(s+\zeta)\, \nu(d\zeta)$*,*

$$A[S_\nu y](t) = \int e^{i\omega\zeta}\, A[y](t+\zeta)\, \nu(d\zeta).$$

*Proof.* (a) Differentiate under the integral. (b) Fubini, then substitute $s \mapsto s+\zeta$ in the inner integral. ☐

*Decomposition of the remainder.* Applying the recursion inside (6) at $t+L$ and substituting $s \mapsto s+L$ (using $e^{-i\omega L} = 1$),

$$A(t+L) = \frac{1}{L} \int E_{\nu^{(s+L)}}\big[y(s+\zeta)\big]\, e^{-i\omega s} \chi\big(\tfrac{s-t}{L}\big)\, ds \;+\; \varepsilon_{\text{tail}}(t),$$

where $\nu^{(s+L)}$ is the law of $\log(J_{e^{s+L}}/(qe^{s+L}))$. Four contributions arise:

(i) $\varepsilon_{\text{tail}}$ (the factor $1/(1+q^n)$ and the indices $j \le M$): super-exponentially small. *Rigorous.*

(ii) the discretisation error (sum versus integral): with a $C^\infty$ compactly supported window, super-polynomially small by Poisson summation. *Rigorous.*

(iii) the main term: replacing $\nu^{(s+L)}$ by the window-centre law $\nu^{(t+L)}$ and applying Lemma 6.3(b) and a Taylor expansion of $A(t+\zeta)$ (legitimate by Lemma 6.3(a)), together with Lemma 6.2, gives $\kappa(t)A(t) + O(e^{-t/2})$. *Rigorous.*

(iv) the residual $\rho(t) = \frac{1}{L} \int e^{-i\omega s} \chi(\frac{s-t}{L}) \int y(s+\zeta)\, (\nu^{(s+L)} - \nu^{(t+L)})(d\zeta)\, ds$, measuring the slow variation of the smoothing kernel across the window.

Thus $r(t) = \varepsilon_{\text{tail}} + \varepsilon_{\text{disc}} + O(e^{-t/2}) + \rho(t)$, and Proposition 6.1 follows once $|\rho(t)| = O(e^{-t/2})$.
*The residual* $\rho$ *as a kernel pairing.* Write $g(s) := \int y(s+\zeta)\, \delta_s(d\zeta)$ with $\delta_s := \nu^{(s+L)} - \nu^{(t+L)}$, so that $\rho(t) = \frac{1}{L} \int g(s)\, e^{-i\omega s} \chi(\frac{s-t}{L})\, ds$, and let $h_\tau$ denote the density of the log-deviation law $\nu^{(\tau)}$. The decisive step moves $y$ outside the oscillatory integral.

**Lemma 6.4** (kernel form of the residual)**.** *One has* $\rho(t) = \dfrac{1}{L} \displaystyle\int_{\mathbb{R}} y(u)\, K(u,t)\, du$*, where*

$$K(u,t) := \int_{\mathbb{R}} e^{-i\omega s}\, \chi\big(\tfrac{s-t}{L}\big)\, \big[h_{s+L} - h_{t+L}\big](u-s)\, ds,$$

*and therefore* $|\rho(t)| \le \frac{1}{L}\, \|y\|_\infty\, \|K(\cdot,t)\|_{L^1}$*.*

*Proof.* Insert $g(s) = \int y(s+\zeta)\, [h_{s+L} - h_{t+L}](\zeta)\, d\zeta$ into the displayed formula for $\rho$, substitute $u = s+\zeta$ in the inner integral, and exchange the order of integration (Fubini: $\chi$ has compact support, $y$ is bounded). The bound is $\big|\int yK\big| \le \|y\|_\infty \int |K|$. ☐

The force of Lemma 6.4 is that the bound holds for *every* bounded $y$: the oscillatory cancellation now resides entirely in $K$ and is no longer entangled with the — possibly large — variation of $y$. The residual is reduced to an $L^1$ estimate for an explicit kernel.

**Proposition 6.5** (smallness of the kernel)**.** *Suppose the log-deviation densities satisfy, uniformly for $\tau$ in a window of width $O(L)$,*

$$\textstyle\int |\zeta|\, h_\tau(\zeta)\, d\zeta = O(e^{-\tau/2}), \tag{7}$$

$$\textstyle\int \big[h_{\tau-\zeta}(\zeta) - h_\tau(\zeta)\big]\, d\zeta = O(e^{-\tau}). \tag{8}$$

*Then $\|K(\cdot,t)\|_{L^1} = O(e^{-t/2})$, hence $\rho(t) = O(e^{-t/2})$ and Proposition 6.1 holds.*

*Proof.* Substituting $\zeta = u - s$ in the definition of $K$,

$$K(u,t) = e^{-i\omega u} \int e^{i\omega\zeta}\, \chi\big(\tfrac{u-\zeta-t}{L}\big)\, \big[h_{u+L-\zeta}(\zeta) - h_{t+L}(\zeta)\big]\, d\zeta.$$

The factor $e^{i\omega\zeta}\chi(\frac{u-\zeta-t}{L})$ is smooth; on the effective support of the densities it equals $\chi(\frac{u-t}{L}) + \theta(u,\zeta)$ with $|\theta(u,\zeta)| \le C|\zeta|$. Hence $K(u,t) = e^{-i\omega u}\big[\chi(\frac{u-t}{L})\, m(u,t) + E(u,t)\big]$ with

$$m(u,t) = \int \big[h_{u+L-\zeta}(\zeta) - h_{t+L}(\zeta)\big] d\zeta, \quad E(u,t) = \int \theta(u,\zeta)\big[h_{u+L-\zeta}(\zeta) - h_{t+L}(\zeta)\big] d\zeta.$$

Since $h_{u+L}$ and $h_{t+L}$ are probability densities, $\int[h_{u+L} - h_{t+L}] = 0$, so $m(u,t) = \int[h_{u+L-\zeta}(\zeta) - h_{u+L}(\zeta)]\, d\zeta = O(e^{-t})$ by (8) at $\tau = u + L$ (and $u - t = O(L)$ on $\operatorname{supp}\chi$). And $|E(u,t)| \le C\int |\zeta|\,(h_{u+L-\zeta} + h_{t+L})(\zeta)\, d\zeta = O(e^{-t/2})$ by (7). As $m$ and $E$ are supported in $u - t = O(L)$,

$$\|K(\cdot,t)\|_{L^1} \le \|\chi\|_{L^1} L \cdot O(e^{-t}) + O(L) \cdot O(e^{-t/2}) = O(e^{-t/2}).$$

The remaining claims are Lemma 6.4 and the decomposition of $r(t)$. □

**▶ OPEN GAP — Gap C — the local limit theorem for the kernel. Work continues here.**

Proposition 6.5 reduces the propagation estimate to two properties of the binomial log-deviation density. The bound (7), a first absolute moment, is elementary (a binomial moment bound with a Hoeffding tail estimate). The *mass-stability* (8), $\int[h_{\tau-\zeta}(\zeta) - h_\tau(\zeta)]\, d\zeta = O(e^{-\tau})$, expresses the scale-regularity of the density; it is a local central limit theorem with one Edgeworth correction for $\mathrm{Bin}(n,q)$ — classical in spirit but not carried out here. One technical layer accompanies it: $\nu^{(n)}$ is supported on a lattice, so $h_\tau$ is the atom-weight profile and (7)–(8) are the lattice form of the local limit theorem — equivalently, Lemma 6.4 is read as a discrete sum. This is where work continues.

### 6.4 From propagation to non-convergence

**Theorem 6.6** (renormalisation)**.** *Let $(x_n)$ be a bounded solution of the strategy-*ALL *recursion for $n > M$. Then for each $t_0$ the sequence $A(t_0 + kL)$ converges, as $k \to \infty$, to a limit $A_\infty$. Moreover: if $x_n$ converges then $A_\infty = 0$; and if $A_\infty \neq 0$ then $(x_n)$ is not eventually monotone.*

*Proof.* Write $A_k = A(t_0{+}kL)$, $\kappa_k = \kappa(t_0{+}kL)$, $r_k = r(t_0{+}kL)$. By Proposition 6.1, $\sum_k |\kappa_k - 1| < \infty$ and $\sum_k |r_k| < \infty$ (geometric series in $e^{-L} < 1$). Solving $A_{k+1} = \kappa_k A_k + r_k$,

$$A_k = \Big(\prod_{j<k} \kappa_j\Big) A_0 + \sum_{j<k}\Big(\prod_{j<i<k} \kappa_i\Big) r_j \;\longrightarrow\; A_\infty,$$

the products and series converging absolutely. If $x_n$ converges then $A(t) \to 0$, so $A_\infty = 0$. If $A_\infty \neq 0$ then $A_k \not\to 0$, so $x_n$ does not converge; a bounded non-convergent sequence is not eventually monotone. $\square$

# 7 The base case and the proof of Theorem 1.1

Theorem 6.6 reduces Step B to a single non-degeneracy: the limit $A_\infty$ must be non-zero.

**Proposition 7.1** (base case)**.** *For every transferred sequence $(x_n)$ of Lemma 5.2, $A_\infty \neq 0$.*

By linearity $A_\infty = A_\infty(b) + A_\infty(g)$, where $g = x - b$ is a non-negative solution of the homogeneous recursion. Here $A_\infty(b)$ is the fundamental Fourier coefficient $\gamma_1(p)$ of the profile $\Phi_p$ of Theorem 1.2, which is non-zero by $\zeta(1 + it) \neq 0$; the modulation by $g$ does not cancel it.

**▶ OPEN GAP — Gap D — the non-degeneracy. Work continues here.**

$A_\infty \neq 0$ is the genuine arithmetic heart of the proof. For the pure strategy All it is $\gamma_1(p) \neq 0$, which holds within the Mellin framework (Gaps A, B) via the non-vanishing of $\zeta$ on the line $\Re s = 1$. For a transferred sequence one must further exclude that the perturbation $g$ cancels the oscillation; this reduces to the linear independence, modulo constants, of the profiles of a family of auxiliary functions $B_r$. Work continues here.

*Proof of Theorem 1.1.* Fix $p \in (0, \frac{1}{2})$. By Lemma 5.1 it suffices to show that $(w_{n,p})$ is neither eventually non-decreasing nor eventually non-increasing.

*Not eventually non-decreasing* is Proposition 4.1.

*Not eventually non-increasing.* Suppose, for contradiction, $w_n \geq w_{n+1}$ for all $n \geq M$. By Lemma 5.2 the transferred sequence $(x_n)$ satisfies $x_n = w_n$ for $n \geq M$ and solves the strategy-All recursion for $n > M$; being a tail of a monotone bounded sequence, $x_n$ converges. By Theorem 6.6 then $A_\infty = 0$, contradicting Proposition 7.1. Hence no non-increasing tail exists.

Both one-sided statements hold, so $(w_{n,p})$ is not eventually monotone; Lemma 5.1 gives infinitely many strict local maxima and minima. $\square$

**Remark 7.2** (strictness)**.** Lemma 5.1 delivers *strict* extrema once $w_{n,p} \neq w_{n+1,p}$ for all large $n$, which holds in all computed data and is expected for every $p \in (0, \frac{1}{2})$; absent it, Theorem 1.1 holds with the non-strict notion of local extremum.

# 8 Numerical evidence

The two predictions have been checked numerically in arbitrary-precision arithmetic (`simulation/b_asymptotics.p` `simulation/renorm_validation.py`). For strategy All the log-periodic oscillation of Theorem 1.2 is confirmed across fourteen orders of magnitude in its amplitude. For the renormalisation argument: the windowed coefficient $A(t_0 + kL)$ is observed to converge to a non-zero limit $A_\infty$ for every tested $p \in \{0.35, 0.40, 0.42, 0.45\}$, and $n\,|\kappa(n) - 1|$ converges to the constant $\frac{p}{2q}|\omega^2 + i\omega|$ of Lemma 6.2 — a direct quantitative confirmation of Proposition 6.1.

# 9 Status and open problems

The skeleton of the proof is complete and the rigorous backbone is substantial: the reduction (Lemma 5.1), the elementary Step A (Proposition 4.1), the exact transfer lemma (Lemma 5.2), the windowed-smoothing identities (Lemma 6.3), the factor estimate (Lemma 6.2), three of the four contributions to the propagation remainder, and the renormalisation theorem (Theorem 6.6). Four steps remain open; they are flagged in the text and summarised here.

| **Gap** | **What is missing** | **Where** |
|---|---|---|
| A | Hayman regularity / sector control of $B(z)$ | §3 |
| B | Mellin contour justification | §3 |
| C | the remainder bound $\lvert\rho(t)\rvert = O(e^{-t/2})$ | §6 |
| D | the non-degeneracy $A_\infty \neq 0$ | §7 |

A complete proof of Theorem 1.1 consists of Proposition 4.1 (done) together with Gaps C and D; Gaps A and B underpin Gap D and the quantitative form of Theorem 1.2. The natural order of attack is C, then A and B, then D. Within Gap C the reduction is complete: Lemma 6.4 and Proposition 6.5 reduce the propagation estimate to two density estimates, (7)–(8), for the binomial log-deviation — a local limit theorem with one Edgeworth correction, in lattice form. That is the current front of work.

## References


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